Ridge Regression: Five Key Points

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Where this fits: Pipeline map step 8 (Model). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
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- **Builds on:** Ridge regression.
- **Compare with:** Lasso regression; L1 and L2 regularisation in neural networks.

1. Overview

Key point: As λ grows, Ridge shrinks the coefficients towards 0 (the largest first) without making any exactly 0, and trades more bias for less **variance** (G-2073). It helps most where least squares is unstable: many or correlated features, or few observations.

The last three Notes built **Ridge regression** (G-1691): the idea, the formulas, and gradient descent. Ridge adds a penalty on the size of the coefficients; its strength is λ (lambda, a number from 0 upwards; the larger λ, the stronger the penalty). This Note collects the five points that are easiest to mix up afterwards. The five points are also common interview questions.

#	Question	Short answer
1	What happens to the coefficients as λ grows?	they shrink towards 0, but never reach it
2	Which coefficients are affected most?	the largest ones
3	What happens to bias and variance?	bias rises, variance falls
4	What happens to the loss function?	its lowest point moves towards 0
5	When should Ridge be used?	when least squares is unstable: many or correlated features, or few observations

A **feature** (G-772) is an input variable (one column of the data table), an **observation** (G-1374) is one record (one row), and the **target** (G-1949) is the value we predict. All diabetes examples below use test size 0.2 and random state 2.

2. Point 1: coefficients shrink but never reach 0

Key point: As λ grows from 0 towards infinity, the coefficients end up close to 0. Ridge does not set any of them to exactly 0.

λ can be any number from 0 upwards. At $\lambda = 0$ there is no penalty, so Ridge is plain linear regression. Figure 1 trains Ridge on the diabetes data for four values of alpha (scikit-learn's

name for λ). How to read it:

- each bar is the coefficient of one of the 10 **features** (G-772), and its height is the coefficient's value (blue positive, red negative);
- note that each panel has its own vertical scale, and its title prints the largest coefficient, so compare those numbers from panel to panel, not the bar heights;
- the table below also gives the test R^2 (the share of the **target** (G-1949)'s spread that the model explains on the test data; see reading R^2).

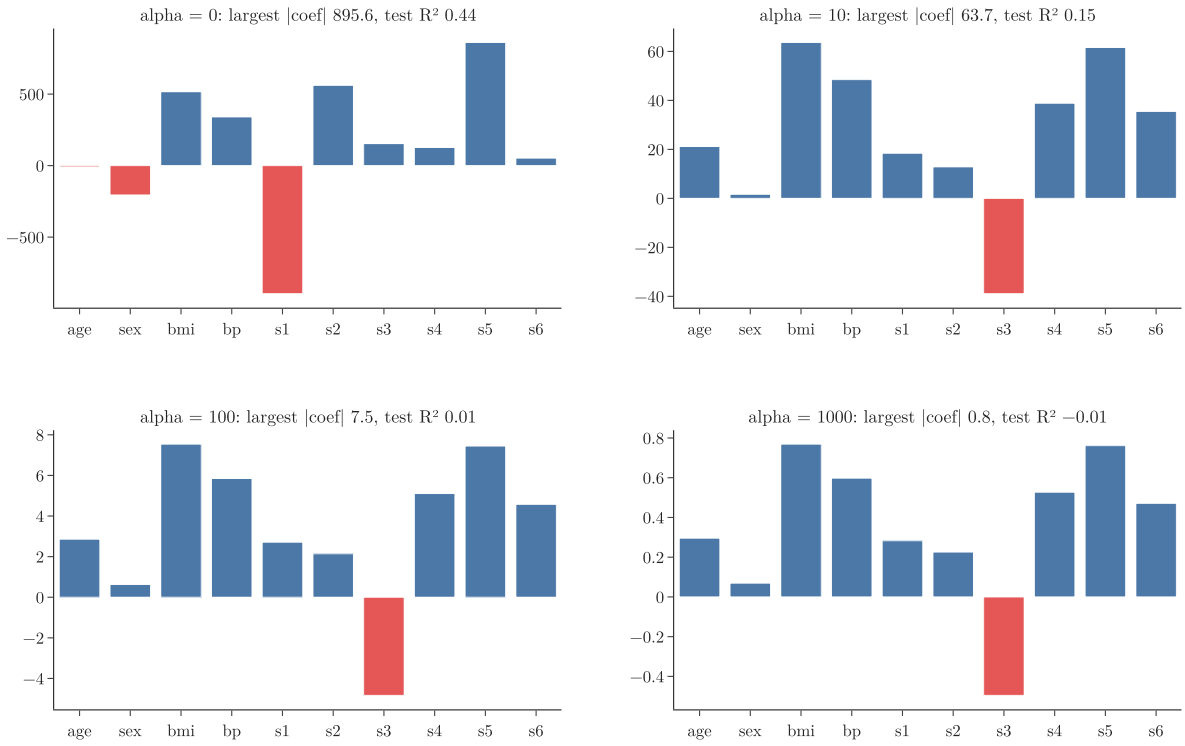


Figure 1: Diabetes coefficients for alpha 0, 10, 100 and 1000; each panel has its own scale

alpha	Largest coefficient (size)	Test R^2
0	895.6	0.44
0.1	485.5	0.45
10	63.7	0.15
100	7.5	0.01
1000	0.8	−0.01
10,000	0.08	−0.01

A small alpha (0.1) shrinks the coefficients and nudges test R^2 up; the larger alphas in the table are there to show shrinkage, and they underfit (Point 3). From alpha 10 on, each tenfold increase makes the coefficients about ten times smaller. The reason is the Ridge answer:

$$\beta = (X^\mathsf{T}X + \lambda I)^{-1}X^\mathsf{T}y$$

Once λ is much bigger than the numbers in $X^\mathsf{T}X$, adding $X^\mathsf{T}X$ to λI changes almost nothing, so:

$$\beta \approx X^\mathsf{T}y/\lambda$$

Ten times the λ then gives a tenth of the coefficients. Here "much bigger" means much bigger than about 3, the largest **eigenvalue** (G-665) of $X^\mathsf{T}X$: a number that says how strongly the matrix stretches its most-stretched direction (see the vectors that do not turn). A one-feature check with made-up numbers $X^\mathsf{T}X = 3$ and $X^\mathsf{T}y = 6$, where the Ridge answer is $6/(3+\lambda)$ and the shortcut is $6/\lambda$:

λ	Ridge answer $6/(3+\lambda)$	Shortcut $6/\lambda$
0	2	(not defined)
30	0.18	0.2
300	0.0198	0.02

At $\lambda = 300$ the two agree to two digits, and ten times the λ gives a tenth of the answer. But even at alpha 10,000, every coefficient is still a small non-zero number. Ridge shrinks coefficients towards 0 but does not set any of them exactly to 0 (ISL §6.2.2). On the way, a coefficient can cross 0 when it changes sign, as s1 and s2 do in the next table, but it does not stay there.

The reason is the slope formula from why the slope shrinks:

$$m = \frac{\text{top}}{\text{bottom} + \lambda}$$

Adding λ to the bottom makes the fraction smaller, but a fraction with a non-zero top never becomes 0.

> **Extra:** Never reaching 0 is the key difference from **Lasso** (G-1047; see the slope reaches exactly 0), which can set coefficients exactly to 0 and so removes features from the model.

3. Point 2: the largest coefficients shrink most

> **Key point:** As λ starts to grow, a large coefficient falls quickly; a small one hardly changes at first.

Figure 2 follows each diabetes coefficient as alpha goes from 0 to 2: its **coefficient path** (G-409). Each line is one **feature** (G-772)'s coefficient: across is alpha, up is the coefficient's value. A line that bends towards the horizontal 0 line is being shrunk.

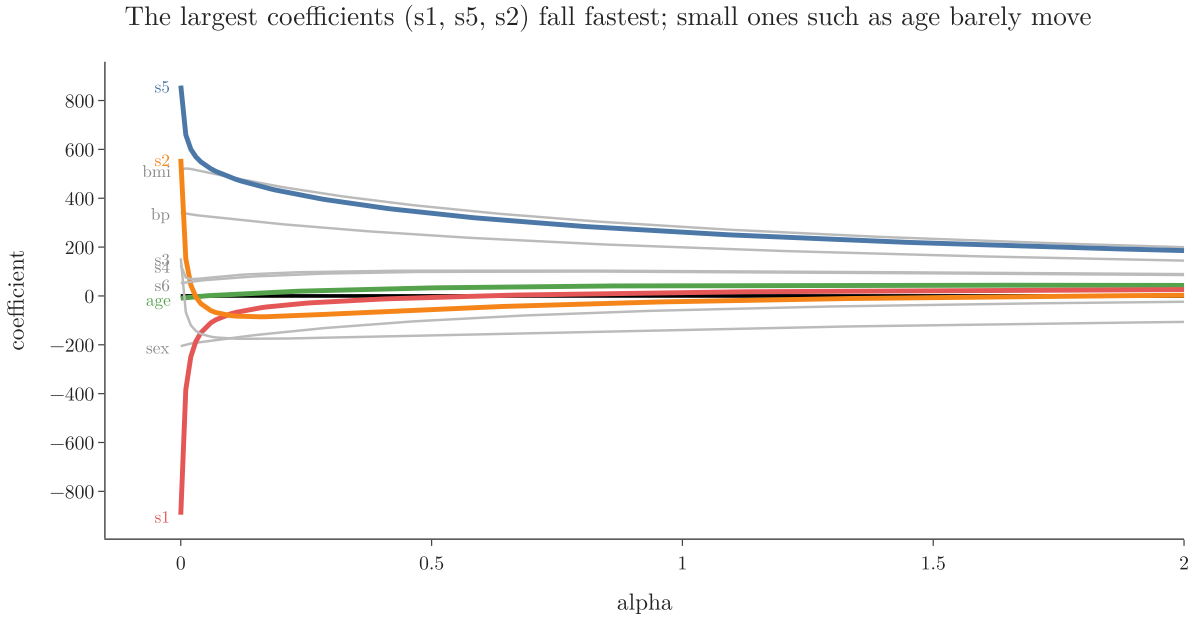


Figure 2: Coefficient paths for alpha from 0 to 2

Feature	alpha 0	alpha 0.01	alpha 0.1	alpha 1
s1	−895.6	−383.7	−72.9	14.4
s5	861.1	659.9	484.4	260.1
s2	561.2	152.7	−80.6	−22.6
age	−9.2	−6.4	6.6	42.2

- **s1, s5 and s2**, the largest, drop steeply even for alpha 0.01.
- **age**, the smallest, barely moves at first.

The penalty is $\lambda \sum \beta_j^2$, and the square grows fast: a coefficient of 900 adds 810,000λ to the loss, while one of 9 adds only 81λ. So reducing the big coefficients lowers the loss far more, and that is where Ridge cuts first.

> **Extra:** A small coefficient can even grow for a while, as age does here (from −9 to 42 at alpha 1). The intuition: s1, s2 and s5 had huge opposing coefficients, and as Ridge cuts them back, part of their effect moves to other features that are mildly correlated with them, age among them (r about 0.3). A quick check agrees: dropping s1, s2 and s5 altogether moves the plain linear regression coefficient of age from −9 to +34. Once alpha is large enough, all coefficients head to 0 together.

4. Point 3: bias rises, variance falls

> **Key point:** A small λ gives low bias and high variance (overfitting); a large λ gives high bias and low variance (underfitting). The best λ is in between.

Recall **bias** (G-287) and **variance** (G-2073): **bias** (G-287) is error from a model too simple to follow the pattern, and **variance** is how much the model changes from one training sample to another. Think of a dartboard: bias is how far the average throw lands from the bullseye, variance is how scattered the throws are.

Measuring bias needs the true curve, so here we use made-up data where we know it: the curve

$$y = 0.7x^2 - 2x + 3$$

plus noise (random scatter with standard deviation 2). We fit a degree-15 polynomial with Ridge to 15 training **observations** (G-1374), then repeat with fresh noise 300 times. Bias^2 compares the average fitted curve with the true curve; variance measures how much the 300 fits spread around their average. Both are measured at 14 test points between the training ones. The expected test error is the sum of bias^2, variance and the noise that cannot be removed.

How to read the figure below:

- alpha runs across and the error runs up, both on log scales (each gridline is ten times the one before);
- orange is the variance, red is bias^2, blue is the expected test error;
- the lowest point of the blue curve is the best alpha.

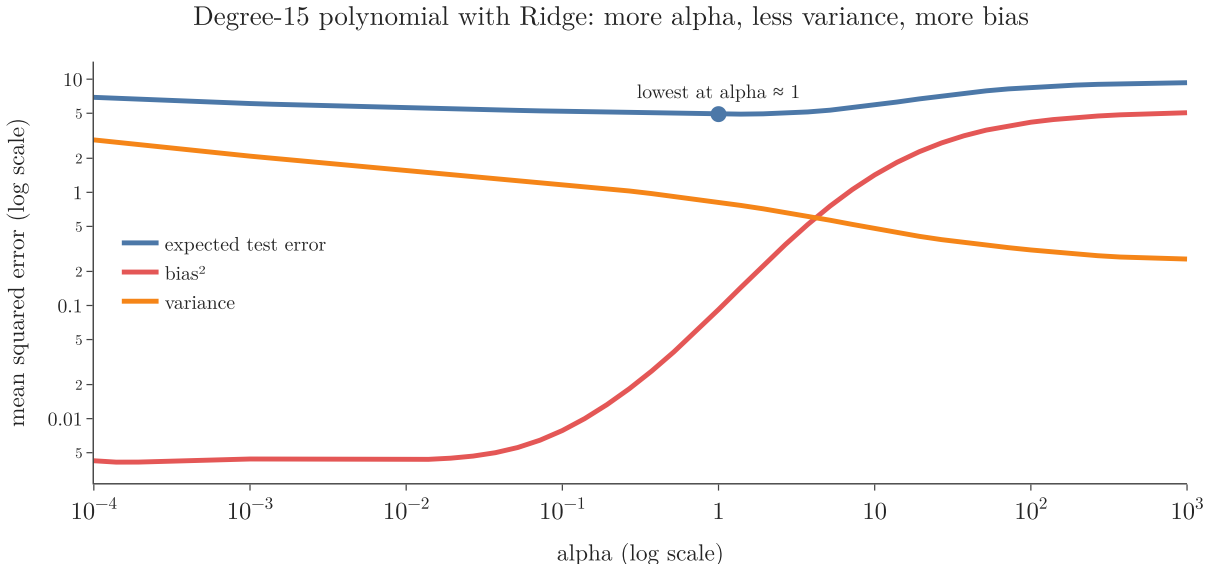


Figure 3: Bias^2, variance and expected test error against alpha

alpha	Bias^2	Variance	Expected test error
0.0001	0.004	2.91	6.91
0.01	0.004	1.58	5.58
1	0.09	0.82	4.91
10	1.43	0.48	5.90
100	4.17	0.31	8.48
1000	5.05	0.26	9.31

- **Small alpha:** the flexible curve bends with each noisy sample, so the variance is high: **overfitting** (G-1429).
- **Large alpha:** the coefficients are squeezed so hard that the curve is too stiff, and bias takes over: **underfitting** (G-2035).
- **In between** (here about alpha 1): the total error is smallest.

So, as in the trade-off, λ moves a model along the **bias-variance trade-off** (G-288; ISL §6.2.1). Choose λ where variance has fallen a lot but bias has not yet risen much.

> **Extra:** Expected test error = bias^2 + variance + noise (ISL §2.2.2, eq. 2.7), and the noise part is $2^2 = 4$ here, the floor no model can beat. The `mlxtend` library's `bias_variance_decomp` function does this repeated-fit calculation for any scikit-learn model. The **features** (G-772) are standardised after `PolynomialFeatures`, as the effect of λ on a flexible model advises. The training x values are fixed and only the noise is redrawn, so every fit covers the same range; test points sit between training points, so no fit has to extrapolate.

5. Point 4: the loss curve rises and its lowest point moves to 0

> **Key point:** Adding λm^2 lifts the loss most where m is large, so the bottom of the curve slides towards m = 0.

To see this in one picture, take the one-feature example and hold the intercept at -2.29, its linear regression value. Then the loss depends only on the slope m:

$$L(m) = \sum_{i=1}^{n} (y_i - mx_i + 2.29)^2 + \lambda m^2$$

Build the curve one point at a time for $\lambda = 50$. For each slope, compute the squared error, add the penalty $50m^2$, and plot the sum:

Slope m	Squared error	Penalty $50m^2$	Loss (sum)
10	56,053	5,000	61,053
17.7	37,285	15,664	**52,950**
27.8	28,342	38,642	66,984

The squared error alone is lowest at 27.8, the linear regression slope. But there the penalty is large, 38,642. At 17.7 the squared error is higher, yet the penalty is so much smaller that the sum is the lowest of all. The penalty has moved the lowest point from 27.8 to 17.7 (StatQuest, "Ridge vs Lasso Regression, Visualized!!!", builds the curve the same way).

Figure 4 draws this curve while λ grows from 0 to 100.

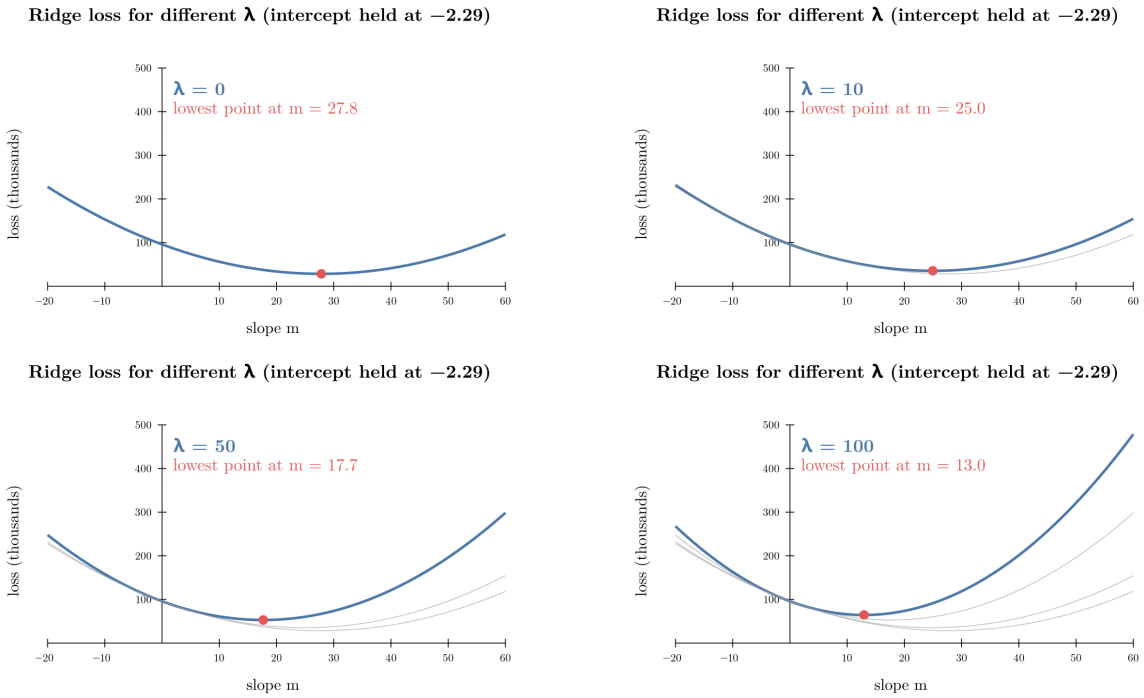


Figure 4: The loss against the slope for $\lambda = 0$, 10, 50 and 100

λ	Slope at the lowest point
0	27.8
10	25.0
50	17.7
100	13.0

The added term λm^2 is 0 at $m = 0$ and grows fast away from it. So the curve rises and narrows, more on the far side from 0, and its lowest point moves towards $m = 0$. Whatever method finds that lowest point (the formula or gradient descent) therefore returns a smaller slope.

5.1 With two coefficients: why "Ridge"?

> **Key point:** With two coefficients, the answer lies where the error contours first touch a circle around the origin.

The next figure is a picture of the squared error over two coefficients, so first the surface it comes from. Take a squared error that is lowest at the OLS answer $\beta = (3, 2.2)$. Written as the extra error above that minimum, it is

$$f(\beta_1, \beta_2) = u^2 + 4v^2$$

where u and v are the distances from the OLS answer along two tilted directions (the ellipse axes). For example, $f(3, 2.2) = 0$ at the OLS answer and $f(0, 0) = 14.3$ at the origin. Think of (β_1, β_2) as a point on the floor and f as the height above it: the surface is a smooth bowl, stretched along one direction, with its lowest point at the OLS answer. Figure 5 draws the bowl. The same bowl seen from above is a contour map: each line joins points of the same height, lines close together mean a steep slope, and the centre of the rings is the lowest point. The orange circle is the penalty (explained below) and the red dot the Ridge answer.

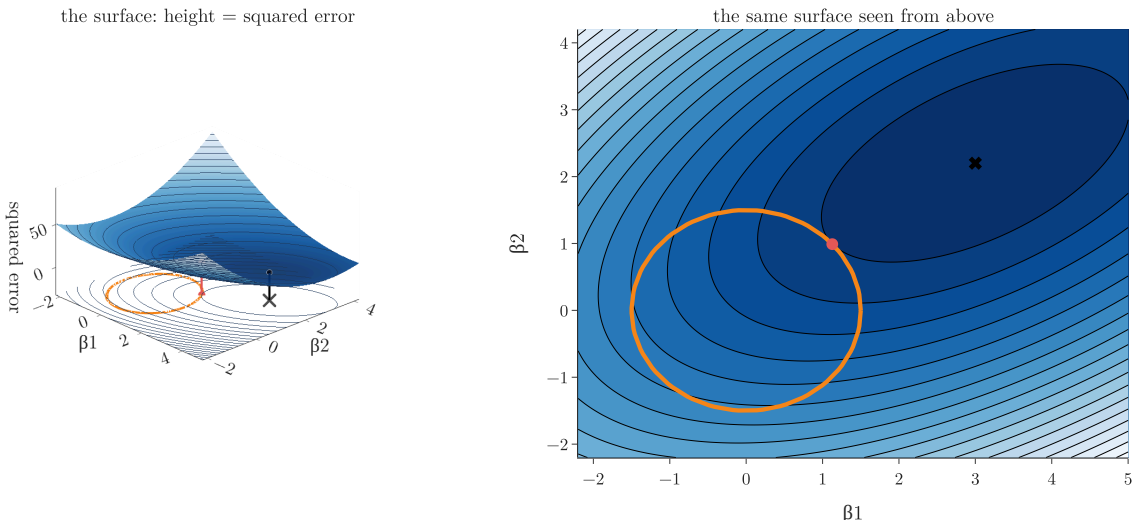


Figure 5: The squared-error bowl over two coefficients (left) and the same bowl seen from above (right). Black cross: the OLS answer (3, 2.2). Orange circle: points with the same penalty. Red dot: the Ridge answer, where the circle touches the lowest contour it reaches.

With two coefficients β_1 and β_2, the loss has two parts: the squared error and the penalty $\lambda(\beta_1^2 + \beta_2^2)$. The squared error alone is lowest at the OLS answer; its contours are ellipses around it (Figure 6, the right panel of Figure 5 without the surface).

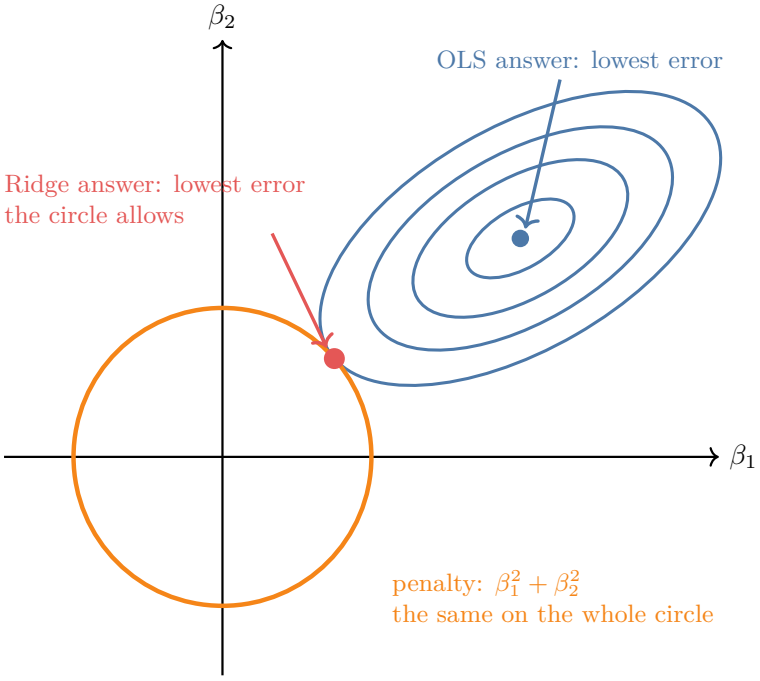


Figure 6: The squared-error contours and the penalty circle

The penalty is the same at every point of a circle around the origin. Ridge looks for the point with the lowest squared error for a given penalty: the point where the ellipses first touch the circle. The touching point is always closer to the origin than the OLS answer: the coefficients as a whole are smaller. In Figure 6 both coefficients shrink; with correlated **features** (G-772) one coefficient can still grow, as age did in Point 2. Read Figure 6 in three steps: (1) every point on one ellipse has the same squared error, and the ellipses shrink towards the OLS answer at their centre; (2) every point on the circle has the same penalty (for $\beta = (3, 2)$ the penalty part is 13 and for $(2, 1.5)$ it is 6.25, worked out below); (3) the best compromise is the point where the smallest ellipse that still reaches the circle just touches it.

The two penalty parts of step (2), one per line:

$$3^2 + 2^2 = 13$$

$$2^2 + 1.5^2 = 4 + 2.25 = 6.25$$

A larger λ means a smaller circle. Seeing Ridge as a hard limit on the coefficients is the **constrained form** (G-454) of the problem; a diamond and a circle compares it with Lasso's.

> **Extra:** The answer to "why is it called Ridge?" is often given with this picture: the solution always lies on the edge of the circle. Historically, the name comes from older work: Hoerl had used *ridge analysis* to study curved response surfaces, and the Ridge

formula looked mathematically similar, so the method was labelled "**ridge regression** (G-1691)" (Hoerl and Kennard 1970, §2).

6. Point 5: use Ridge where least squares is unstable

Key point: Ridge helps when the least-squares coefficients change a lot from sample to sample. That happens with many features, with correlated features, and with very few **observations** (G-1374). With one feature and plenty of data there is little to gain.

Regularisation fights **overfitting** (G-1429), and Ridge works best where the least-squares coefficients have high variance (ISL §6.2.1). Three situations cause it:

- **Many features for the number of observations.** Each extra coefficient is one more thing to estimate from the same data.
- **Correlated features.** A large positive coefficient on one can cancel a large negative one on its partner, like the s1 (−896) and s2 (+561) pair above; Ridge's size limit stops this (ESL §3.4.1).
- **Very few observations,** even with one **feature** (G-772). The two-point line of the effect of λ with one feature is this case: its slope swings widely from sample to sample.

A common rule of thumb is "use Ridge with 2 or more features". The rule covers the first two situations, which are the usual ones in practice.

Figure 7 tests the point on the diabetes data with only 40 training observations, averaged over 200 random splits, with alpha chosen by cross-validation (splitting the training part into folds and testing on each in turn) on the training part. With bmi as the only feature, Ridge scores 0.30 against 0.31 for linear regression: nothing to gain, because 40 observations pin down one slope well. With all 10 features, linear regression drops to 0.29 and Ridge lifts it to 0.39.

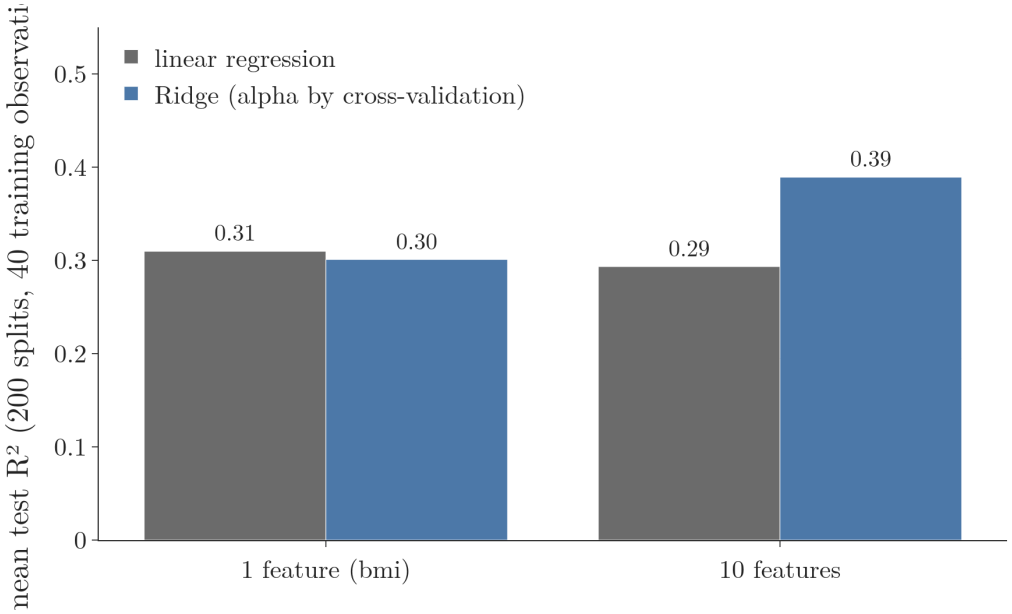


Figure 7: Mean test R^2 over 200 splits with 40 training observations: linear regression (grey) against Ridge with alpha chosen by cross-validation (blue). With one feature Ridge gains nothing (Ridge 0.30 against 0.31 for linear regression); with 10 features it gains 0.10 (0.29 to 0.39).

The third situation shows up when the same one-feature test is run with only 3 training observations. Both models are then poor, but linear regression is far worse: its mean test R^2 is -3.44 against -1.79 for Ridge (script `one_vs_many.py`). From about 6 observations on, the gain is gone.

7. Summary

#	Point	Why
1	As λ grows, the coefficients end up near 0, but Ridge does not set any to exactly 0	because λ only adds to the bottom of the slope fraction, and a fraction with a non-zero top never becomes 0; so Ridge keeps every feature
2	The largest coefficients shrink fastest; small ones barely change at first	because the penalty squares each coefficient, so cutting a big one lowers the loss far more
3	Larger λ: higher bias, lower variance. Pick λ in between	because the expected test error is lowest where variance has fallen a lot but bias has not yet risen much
4	The loss curve rises and its lowest point slides towards 0	because the penalty is 0 at 0 and grows fast away from it, so the formula or gradient descent returns a smaller slope
5	Use Ridge where least squares is unstable: many or correlated features, or very few observations	because there the least-squares coefficients swing from sample to sample; with one feature and plenty of data Ridge gains nothing

Ridge never sets a coefficient to exactly 0. Lasso (one feature: the slope reaches exactly 0) does: its penalty puts a sharp corner in the loss curve at 0. So these five points answer what λ does to a Ridge model and when the trade is worth making.

8. Sources

Built from

- CampusX, "5 Key Points - Ridge Regression | Part 4 | Regularized Linear Models", YouTube, https://www.youtube.com/watch?v=8osKeShYVRQ
- StatQuest with Josh Starmer, "Ridge vs Lasso Regression, Visualized!!!", YouTube, https://www.youtube.com/watch?v=Xm2C_gTAl8c. Building the penalised loss curve point by point (Section 5).

Other references

- **ISL:** James, G., Witten, D., Hastie, T. and Tibshirani, R. *An Introduction to Statistical Learning*, 2nd ed. Springer, 2021. Section 2.2.2, p. 34; Sections 6.2.1–6.2.2, pp. 240–241.
- **ESL:** Hastie, T., Tibshirani, R. and Friedman, J. *The Elements of Statistical Learning*, 2nd ed. Springer, 2009. Section 3.4.1, p. 63.
- **Hoerl and Kennard 1970:** Hoerl, A. E. and Kennard, R. W. "Ridge Regression: Biased Estimation for Nonorthogonal Problems." *Technometrics* 12(1), 55–67, 1970. Section 2.

9. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Coefficient path (G-409)	How each coefficient changes as the regularisation strength grows, drawn as one line per feature, so we can see which coefficients are shrunk towards 0.
Constrained form (G-454)	Writing regularisation as: make the loss as small as possible while the coefficients stay inside a fixed budget (a circle for Ridge). The picture shows how the penalty limits the size of the coefficients.
Variance (of a model) (G-2073)	How much a model's predictions change when it is trained on a different sample of the data; a different meaning from the variance of a column.
Ridge regression (G-1691)	Linear regression with a penalty on the sum of squared coefficients added to the loss; the penalty keeps the coefficients small, which reduces overfitting (L2 regularisation).
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Observation (G-1374)	One record of a dataset: one row of the data table.
Target (G-1949)	The output column we predict, such as the class; also called the label.
Eigenvalue (G-665)	The number by which a matrix stretches or shrinks one of its eigenvectors: $A\mathbf{v} = \lambda\mathbf{v}$. In PCA, the eigenvalues of the covariance matrix are the variances along the principal components.

Term	Meaning
Lasso regression (G-1047)	Linear regression with a penalty on the sum of absolute coefficients (L1); it shrinks coefficients and can set some exactly to 0, which removes those features.
Bias (of a model) (G-287)	Error from a model being too simple to capture the true relationship.
Overfitting (G-1429)	Learning the training data too closely, noise included; fails on new data.
Underfitting (G-2035)	Being too simple to capture the pattern; fails on all data.
Bias-variance trade-off (G-288)	Lowering bias by adding complexity tends to raise variance, and the reverse.